Application of CNN-BiLSTM-PINN integrating physical and numerical constraints in AVO LMR inversion

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Abstract

A convolutional neural network-bidirectional long short-term memory-physics-informed neural network (CNN-BiLSTM-PINN) architecture is proposed that integrates physical and numerical constraints for seismic elastic parameter inversion. Traditional seismic inversion methods often face challenges of insufficient accuracy or low computational efficiency when dealing with complex geologic conditions. To address these issues, two network architectures — convolutional neural network-physics-informed neural network (CNN-PINN) and CNN-BiLSTM-PINN — were constructed, and a hybrid strategy combining model-driven and data-driven approaches was designed. Systematic experiments using the Marmousi2 model compared the inversion performance across different network architectures and driving strategies. Results indicated that under the hybrid driving mode (model 0.3 + numerical 0.7), CNN-BiLSTM-PINN provided clearer boundaries, better structural continuity, and improved resolution for the Lamé parameter (λ) and shear modulus (μ), whereas CNN-PINN demonstrated statistically significant advantages in density (ρ) inversion ($p < 0.001$). Practical applications in the CO_2 flooding demonstration area of X oilfield further demonstrated that the method accurately delineated the CO_2 flow range and aligned well with well logging data. The study also found that although the CNN-BiLSTM architecture showed higher training efficiency than CNN in supervised learning, its computational complexity increased significantly within the PINN framework, primarily due to the interaction between physical constraint calculations and the recursive structure of the BiLSTM. These findings offer a novel paradigm for seismic elastic parameter inversion, enhancing inversion accuracy and stability while preserving physical reasonability, with significant implications for oil and gas exploration and carbon sequestration monitoring.

Introduction

Seismic prestack amplitude variation with offset (AVO) inversion is an important technique in oil and gas exploration for estimating elastic parameters such as P-wave velocity, S-wave velocity, density, and derived attributes (e.g., Lamé parameters) from amplitude variations with incident angle, enabling reservoir characterization and fluid prediction (Grana et al., 2013). However, it is a nonlinear, ill-posed inverse problem. Traditional approaches often rely on accurate initial models, linearized approximations of the Zoeppritz equation, and iterative optimization, making them sensitive to noise and prone to local minima, with limited resolution in complex geologic settings such as thin interbeds (Muskat and Meres, 1940; Koefoed, 1955).

Recent advances in machine learning and deep learning (DL) have offered new solutions to these challenges. Early studies, such as adaptive network fuzzy inference systems (Karimpouli and Fattahi, 2018) and Boltzmann machines (Phan and Sen, 2019), demonstrated the feasibility of directly predicting elastic parameters from seismic data without initial models. Convolutional neural networks (CNNs) have become widely used in prestack inversion because of their strong spatial feature extraction ability (Biswas et al., 2019; Aleardi and Salusti, 2021; Cui et al., 2021; Liu and Zhong, 2021; Cao et al., 2022; Junhwan et al., 2022; Liu et al., 2023; Sun et al., 2023a; Sun et al., 2023b; Wang and Wang, 2023). Enhancements include physical guidance (Biswas et al., 2019), U-Net architectures (Cui et al., 2021; Cao et al., 2022), synthetic data pretraining (Li et al., 2022; Zhang et al., 2022), sparse constraints (Cao et al., 2022), uncertainty estimation

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(Aleardi, 2022; Junhwan et al., 2022), and targeted designs for challenges like thin interbeds (Wang and Wang, 2023). Recurrent architectures such as long short-term memory (LSTM) networks have also been explored for capturing seismic data's sequential dependencies (Aleardi, 2022).

Although purely numerical-driven DL methods have achieved progress, they remain dependent on large amounts of labeled data and risk producing physically inconsistent results. Physics-informed neural networks (PINNs), introduced by Raissi et al. (2019), address these issues by embedding governing partial differential equations directly into network training, ensuring that predictions satisfy physical laws. Applications in geophysics include PINN-based full-waveform inversion (FWI) in 2D acoustic media (Rasht-Behesht et al., 2022) and probabilistic PINN frameworks for petrophysical inversion with uncertainty quantification (Li et al., 2024). These advances highlight the potential of PINNs in seismic inversion.

Hybrid strategies that combine numerical-driven and physics-driven terms — such as forward operator construction (Biswas et al., 2019), neural simulation of forward processes (Song et al., 2022), and weighted loss designs — have improved inversion accuracy, particularly with sparse labels or strong noise. However, the application of a dual-driven PINN framework to prestack AVO inversion, explicitly combining model-driven (lambda-mu-rho [LMR]-based) and numerical-driven loss terms with tunable weights and systematically comparing different backbone networks (CNN versus convolutional neural network-bidirectional long short-term memory [CNN-BiLSTM]), has not been comprehensively explored. This study proposes such a framework, aiming to optimize the balance between physical consistency and data-fitting capability to achieve high-precision, high-resolution elastic parameter inversion in accordance with physical laws.

Method

PINN architecture

As shown in Figure 1, the CNN-PINN overall architecture adopts a "model + numerical dual-driven" strategy, optimizing the PINN framework by combining physical constraints (model constraints) with numerical constraints. The numerical constraints are provided by a trained forward neural network, which serves the same function as the forward part of the LMR physical model, jointly forming a dual-driven forward mechanism. In the data processing workflow, angle-stacked seismic data are first input and processed through a CNN encoder for multiscale feature extraction. Multiple convolution kernels (kernel 1, kernel 2, kernel 3) represent the input data at different scales, followed by batch normalization, activation, and dropout to extract local semantic features. These features are used for LMR parameter inversion to predict the elastic parameters of the formation.

To optimize the PINN training process, this study assigns weights to physical constraints and numerical constraints in the loss function, ensuring that both drive the forward process together. The LMR formula is used to calculate reflection coefficients, which are convolved with a Ricker wavelet to generate synthetic angle seismic data, providing physical constraints. Meanwhile, the trained forward neural network uses input data to directly generate numerically simulated synthetic angle seismic data as numerical constraints. Finally, the loss function comprehensively considers both constraints and is trained through the Adam optimizer. This enables the network to satisfy physical laws while fully using the numerical advantages of DL models, improving the stability and accuracy of geophysical parameter inversion.

As shown in Figure 2, only the CNN-BiLSTM network component of the CNN-BiLSTM-PINN model is illustrated, highlighting the replacement of the CNN encoder in Figure 1 with a CNN-BiLSTM encoder to enhance the modeling capability of temporal

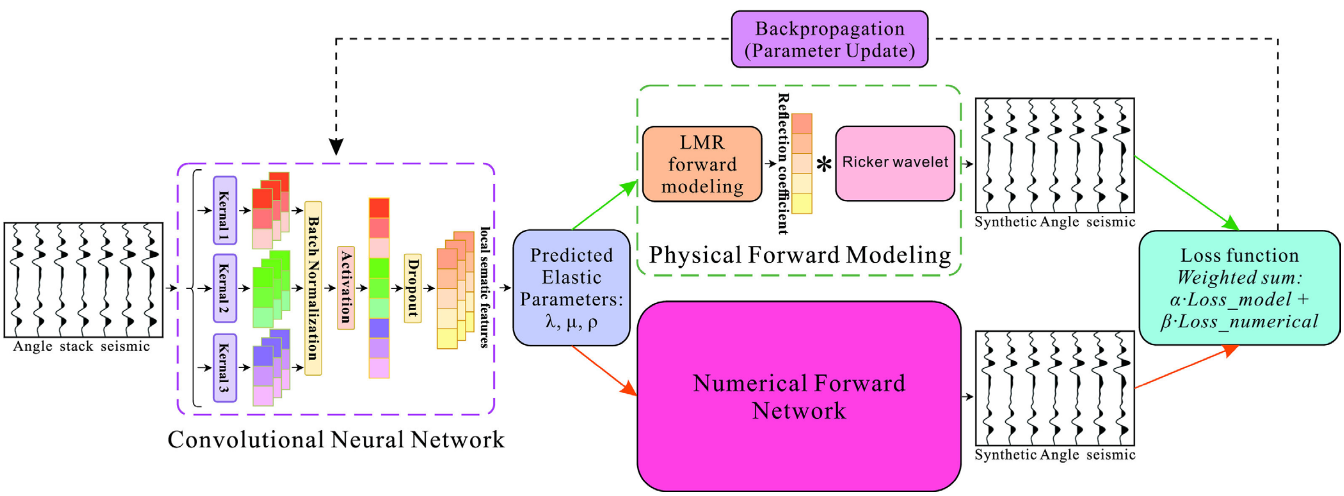


Figure 1. Model + numerical dual-driven CNN-PINN overall architecture diagram.

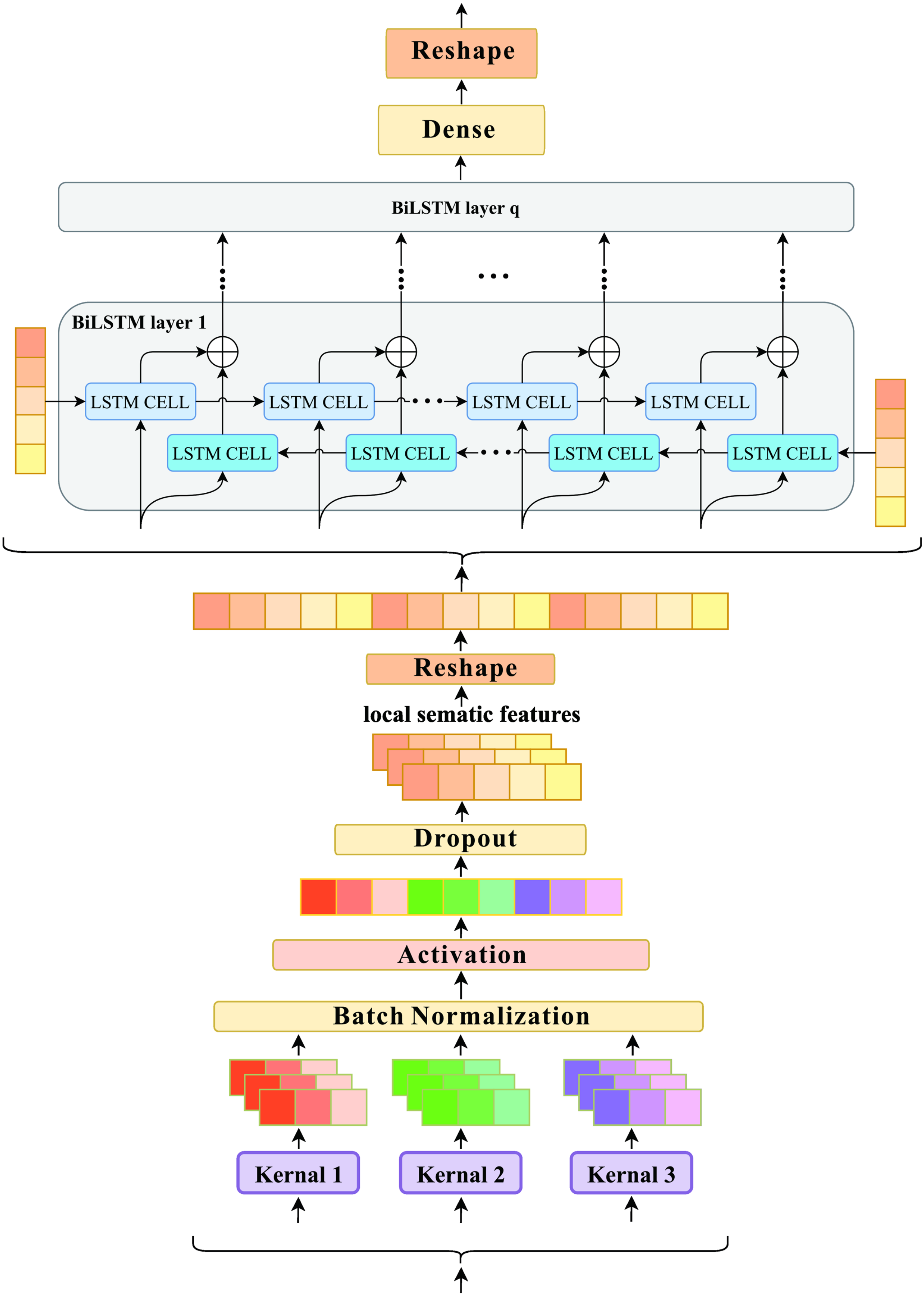


Figure 2. CNN-BiLSTM network architecture for the CNN-BiLSTM-PINN model.

information and further optimize the "model + numerical dual-driven" strategy. In this architecture, the overall forward mechanism remains unchanged, still consisting of the LMR physical model (model constraint) and the trained forward neural network (numerical constraint). First, the CNN performs local feature extraction on angle-stacked seismic data, forming feature sequences after operations such as batch normalization, activation, and dropout. Subsequently, the BiLSTM network processes these feature sequences using forward and backward LSTM structures to capture the spatiotemporal dependencies of seismic data, thereby enhancing the network's ability to represent complex formations. In the forward process, the network simultaneously accepts constraints from the LMR formula and the trained forward neural network to ensure the reliability of the forward mechanism. The loss function assigns weights to these two parts, ensuring that the PINN network adheres to physical laws while being influenced by numerical constraints during the training process, thereby improving the accuracy and stability of geophysical parameter inversion. Finally, the optimization process adopts the Adam algorithm to minimize the loss function and optimize network weights, achieving efficient inversion solutions. Compared with traditional PINN methods, this model can more accurately adapt to nonlinear complex formations, improve the inversion accuracy of reservoir elastic parameters, and reduce dependence on high-quality physical priors with the help of data constraints, thereby enhancing the model's generalization ability.

In contrast to conventional inversion frameworks, the proposed CNN-PINN and CNN-BiLSTM-PINN methods fundamentally differ from purely physics-driven and purely numerical-driven approaches. Traditional FWI relies entirely on numerical simulation (e.g.,solving the wave equation) and classical optimization algorithms such as Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm(LBFGS) to minimize the misfit between observed and synthetic data. It does not involve neural networks and typically uses hyperbolic feature seismic data obtained through forward modeling, which often demands a high-quality initial model and faces challenges in complex geologic settings.

Conversely, our approach embeds physical constraints (e.g., the LMR formulation) into a DL framework, combining the global consistency of physical laws with the local feature learning capability of neural networks. Furthermore, by using prestack angle-domain seismic data and applying near-angle stacking to enhance the signal-to-noise ratio, our method better captures elastic parameter variations in complex structures such as faults and thin interbeds.

Unlike conventional CNN-based inversion, which requires large volumes of labeled data and often suffers from limited generalization and overfitting in structurally complex areas, the CNN-PINN and CNN-BiLSTM-PINN architectures integrate physical priors into the network training. This enables unsupervised inversion, where the outputs not only fit the numerical simulation but also adhere to physical consistency, yielding more accurate, stable, and geologically plausible results.

Physical constraints

For seismic inversion, incorporating physical constraints is vital to enhance the accuracy of the resulting model and its ability to generalize to unseen data. Conventional inversion techniques often place significant reliance on initial models and iterative optimization procedures, whereas PINNs offer an improved approach. PINNs leverage physical principles by integrating prior physical knowledge directly into the network training, thereby promoting better adaptability to geologically complex settings. In this study, the linearized LMR approximation serves as a physical constraint, guiding the inversion process to conform to established physical laws, as detailed in references (Goodway et al., 1997; Russell et al., 2003).

The LMR formulation, used in seismic inversion, provides a linear approximation of seismic reflection coefficients based on subsurface elastic parameters. The primary objective of the LMR approach is to directly estimate the elastic properties of subsurface reservoirs — specifically the Lamé parameter (λ), shear modulus (μ), and density (ρ) — from the reflection behavior observed in seismic data. λ exhibits sensitivity to volumetric strain and displays a strong correlation with the presence of pore fluids, making it useful for fluid discrimination. μ, on the other hand, is indicative of the stiffness of the rock matrix and remains relatively unaffected by fluid content, rendering it a key parameter for lithological characterization. Together, these elastic parameters are essential for identifying reservoir fluids and distinguishing lithologies.

The LMR formulation is derived through a linear approximation of the Aki–Richards equation, with its fundamental expression given by

$$R(\theta) \approx A(\theta) \cdot \Delta\ln(\lambda) + B(\theta) \cdot \Delta\ln(\mu) + C(\theta) \cdot \Delta\ln(\rho), \quad (1)$$

$$A(\theta) = \frac{1}{2}\left(1 + \tan^2(\theta)\right)\frac{\lambda}{\lambda + 2\mu}, \quad (2)$$

$$B(\theta) = \frac{1}{2}\left(1 + \tan^2(\theta)\right)\frac{2\mu}{\lambda + 2\mu} + \frac{1}{2}\left(-4\frac{\lambda}{\lambda + 2\mu}\sin^2(\theta)\right), \quad (3)$$

$$C(\theta) = \frac{1}{2}\left(1 + \tan^2(\theta)\right)\left(-\frac{1}{\rho}\right), \quad (4)$$

where $R(\theta)$ represents the seismic reflection coefficient, whereas $\Delta\ln(\lambda)$, $\Delta\ln(\mu)$, and $\Delta\ln(\rho)$ correspond to the relative changes in λ, μ, and ρ, respectively. $A(\theta)$, $B(\theta)$, and $C(\theta)$ denote angle-dependent coefficients that characterize how the reflection coefficient varies with the angle of incidence (θ). The coefficient $A(\theta)$ primarily reflects the contribution of fluid-sensitive parameters (λ), $B(\theta)$ emphasizes the impact of lithology-related parameters (μ), and $C(\theta)$ is associated with ρ. Through this specific formulation, the LMR approximation effectively separates the influences of fluid and lithology parameters on the reflection coefficients, facilitating a more precise estimation of reservoir elastic properties.

Subsequently, synthetic seismic data in the angle domain are generated by convolving the reflection coefficient $R(\theta)$ with a Ricker wavelet (W):

$$\mathrm{Seis} = R(\theta) * W, \tag{5}$$

where Seis is the observed seismic trace (seismogram), * denotes temporal convolution, $R(\theta)$ is the angle-dependent reflectivity series evaluated at incidence angle θ, and W is the source wavelet.

The LMR formulation used in this study is derived from a linearized approximation of the Aki–Richards equation, which assumes isotropic, weak-contrast media and neglects frequency-dependent dispersion effects. These assumptions are generally valid for prestack seismic data at small to moderate incident angles, making the formulation suitable for the targeted reservoir characterization scenarios in this study. The choice of LMR as the physical constraint allows for a balance between computational efficiency and geologic interpretability, which is critical for validating the proposed hybrid-driven inversion framework.

Numerical constraint

This study uses the Marmousi2 model as the basis for generating training data. The Marmousi2 model is widely applied in the field of seismology due to its ability to provide high-resolution subsurface medium properties, including P-wave velocity, S-wave velocity, and density. To improve computational speed, the original Marmousi2 data were first downsampled to one-tenth of the original data in horizontal and vertical directions:

$$\lambda = \rho\left(V_{\mathrm{P}}^2 - 2V_{\mathrm{S}}^2\right), \tag{6}$$

$$\mu = \rho V_{\mathrm{S}}^2, \tag{7}$$

where V_{P} represents the P-wave velocity, V_{S} the S-wave velocity, and ρ the density.

During the data preparation phase, P-wave velocity, S-wave velocity, and density were converted to λ, μ, and ρ according to equations 6 and 7. Figure 3a–3c displays the Marmousi2 model structure represented by LMR parameters. These model properties established a solid physical foundation for subsequent seismic wave simulation.

To generate synthetic seismic data in the angle domain, this study used the LMR reflection coefficient formula (equations 1–5). We selected three representative reflection angles — 10°, 20°, and 30° — to simulate seismic reflection responses at different angles. This angle range covers common observation angles in actual seismic exploration, aiming to ensure the diversity and geologic realism of the simulated data.

The seismic source was simulated using a Ricker wavelet with a dominant frequency of 25 Hz and a length of 64 sampling points. By convolving this wavelet with the reflection coefficient sequence, the process of seismic waves passing through reflection interfaces was simulated. This forward simulation process ultimately generated synthetic seismic data profiles for near (10°), mid (20°), and far (30°) angles, as shown in Figure 3d–3f. These synthetic data sets were subsequently used as training samples for the model and numerical hybrid-driven CNN-PINN and

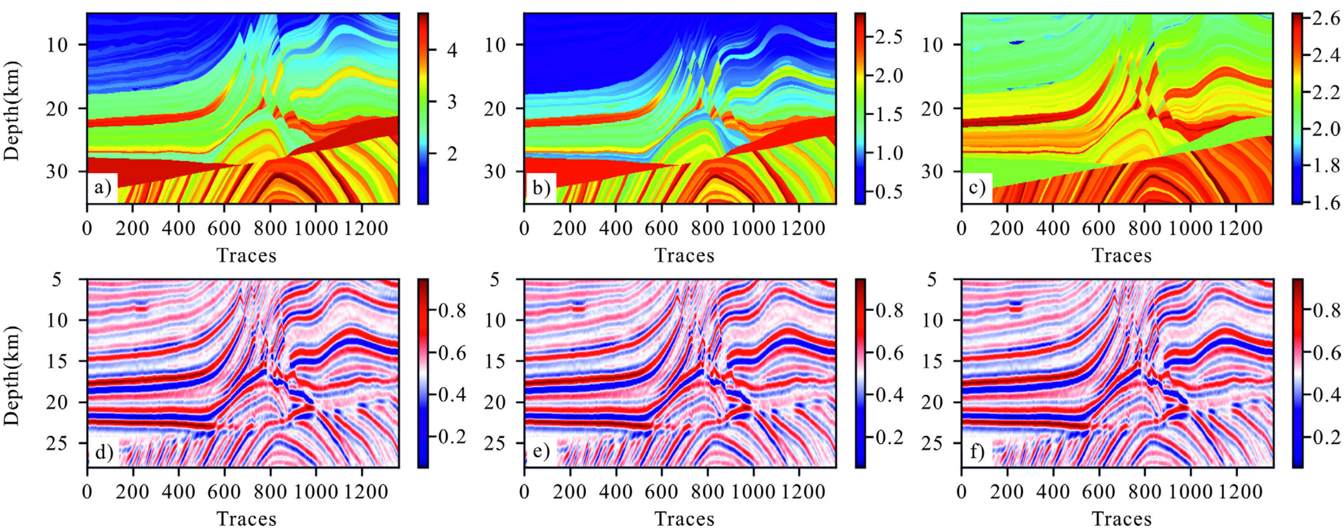


Figure 3. Marmousi2 model and synthetic seismic records (used as observed seismic data): (a-c) λ, μ, and ρ models and (d-f) near, mid, and far-angle seismic records.

CNN-BiLSTM-PINN. In total, the data set consists of 1088 training samples and 273 testing samples (a split ratio of 8:2). Each sample contains three prestack angle gathers (10°, 20°, 30°) of length 230, resulting in an input data shape of (3, 230, 1) and a corresponding output of three inverted elastic parameters: λ, μ, and ρ. By integrating the physical constraints contained in the Marmousi2 model with the LMR calculation framework, the generated training data ensured physical consistency, providing high-quality, high-fidelity learning samples for subsequent inversion tasks.

Figure 4 shows the changes in training loss and validation loss with training epochs for the CNN model (Figure 4a) and CNN-BiLSTM model (Figure 4b) during the numerical training phase. Observing the loss curves, both models demonstrate good convergence characteristics. Training loss and validation loss decrease rapidly in the early stages of training and subsequently tend to stabilize, eventually converging to relatively low levels. Notably, the two loss curves remain closely aligned throughout the training process, without the validation loss being significantly higher than the training loss or showing divergence, indicating that both models have good generalization ability and effectively avoid overfitting. Comparing the convergence processes of the two models: the CNN-BiLSTM model (Figure 4b) not only converges to a lower final loss value (close to zero) than the CNN model (Figure 4a), suggesting that it may have better data fitting ability and prediction accuracy, but also converges faster, with the loss value significantly decreasing and approaching a stable state in fewer epochs (about 50 epochs). Further analysis combined with the training time data provided in Table 1. The CNN model was trained for 300 epochs with a total time of 257.21 s. By contrast, the CNN-BiLSTM model was trained for only 229 epochs (having already achieved excellent convergence), with its total time significantly reduced to 109.02 s, less than half the training time of the CNN model. Overall, in the numerical training task of this study, the CNN-BiLSTM model shows significant advantages over the CNN model. It not only achieves a lower final loss, indicating higher potential prediction accuracy, but also has higher learning efficiency, reaching excellent convergence in fewer training epochs and significantly shorter training time. This suggests that although the CNN-BiLSTM model structure is more complex, its enhanced ability to capture sequence information significantly improves learning efficiency and final performance, making it superior to the standard CNN model in accuracy and efficiency dimensions.

Figure 5 provides a quantitative evaluation and comparison of synthetic seismic data predicted by two DL models: CNN and CNN-BiLSTM. The figure displays seismic data sections with near, mid, and far angle stacks predicted by both models, as well as their respective difference sections compared to the actual synthetic seismic data (Figure 3d–3f). From the macroscopic visual effect of the prediction results (Figure 5a–5c and Figure 5g–5i), both models can effectively predict the main structural morphology and reflection phase axis characteristics of the seismic sections, indicating their ability to learn and reconstruct seismic wavefields. However, by comparing their respective difference sections (i.e., prediction errors or residuals), the differences in prediction

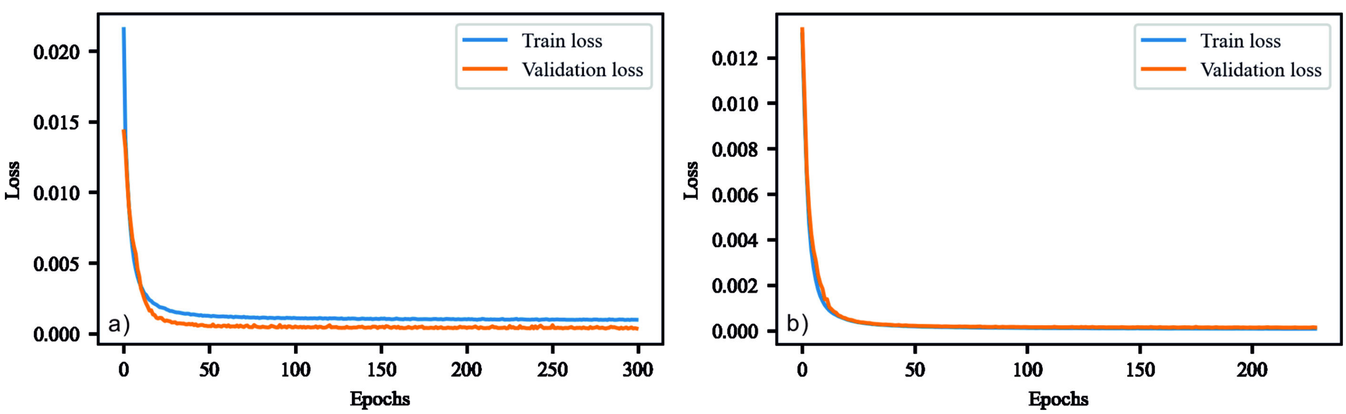


Figure 4. Numerical training curves of (a) CNN and (b) CNN-BiLSTM.

Table 1. Comparison of training time for different methods.

Method/time (s)	CNN	CNN-BiLSTM	CNN-PINN	CNN-BiLSTM-PINN
Model + numerical constraint			288.63 s	468.33 s
Pure model constraint	257.21 s (300 epochs)	109.02 s (229 epochs)	377.67 s	417.2 s
Pure numerical constraint			302.07 s	398.65 s

accuracy between the two models can be more clearly revealed.

The difference sections of the CNN model (Figure 5d–5f) show relatively strong residual energy, especially in structurally complex areas, fault edges, and near strong reflection phase axes, where there are obvious error distributions with certain structural morphologies. This indicates that the standard CNN model has limitations in accurately predicting seismic response details in these areas, and there are still nonnegligible systematic deviations between its prediction results and the actual data.

In contrast, the difference sections of the CNN-BiLSTM model (Figure 5l–5n) exhibit significantly reduced residual energy levels. The error distribution across the entire section is more random, generally smaller in amplitude, and lacks obvious systematic error structures related to geologicespecially comparing Figure structures. This clearly indicates that the CNN-BiLSTM model can more accurately predict synthetic seismic data, with higher consistency between its prediction results and the actual data. This performance improvement may be attributed to the BiLSTM layer's ability to effectively capture temporal and spatial sequence dependencies within and between seismic traces, thereby modeling wavefield characteristics more accurately.

Through comprehensive comparative analysis, although CNN and CNN-BiLSTM models can predict the main features of synthetic seismic data, the CNN-BiLSTM model demonstrates clear advantages in prediction accuracy. Its prediction results have smaller residuals compared to the actual synthetic data and are closer to random noise, proving that the CNN-BiLSTM architecture has stronger fitting capability and higher fidelity in seismic data prediction tasks.

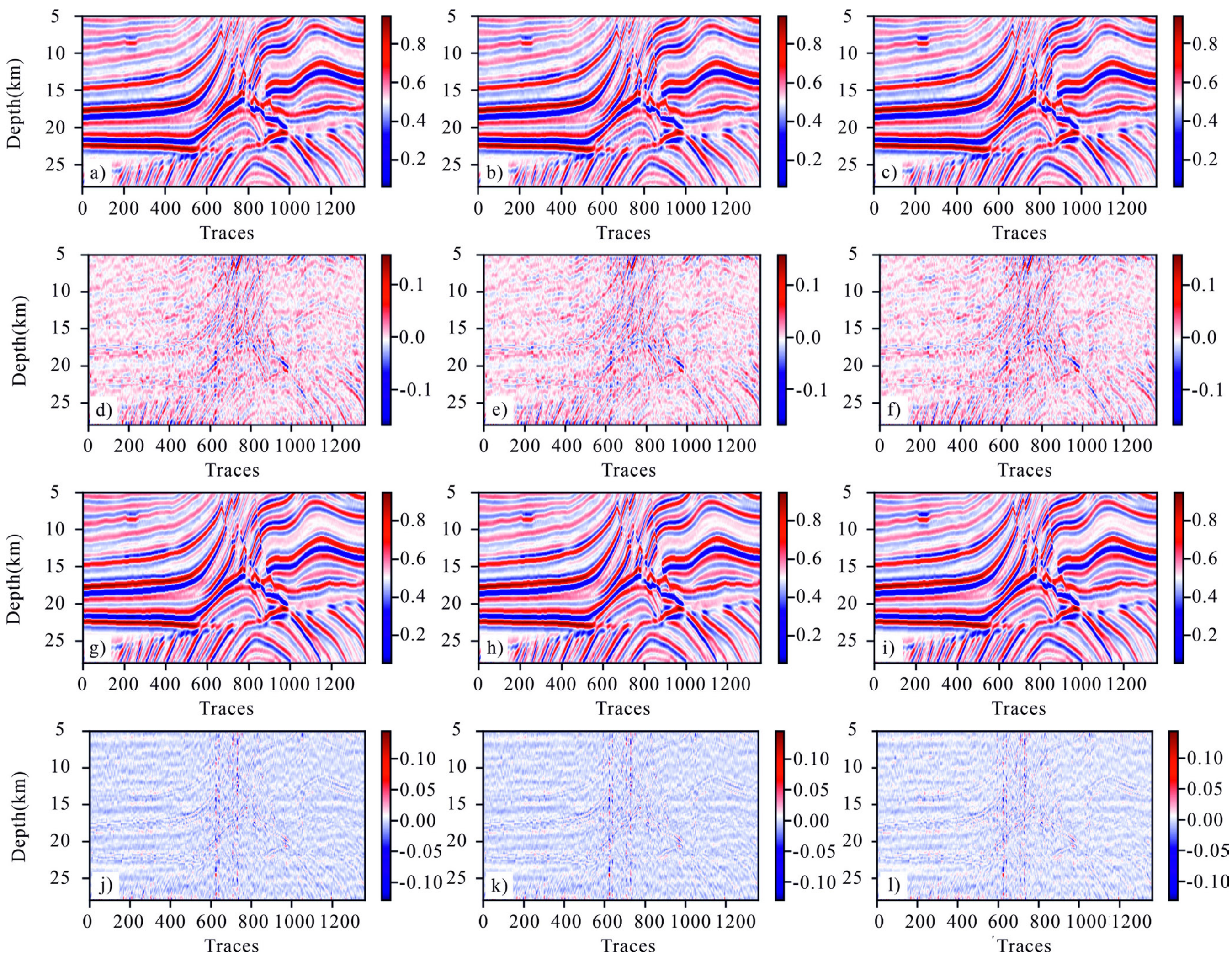


Figure 5. Comparison of synthetic seismic data predicted by CNN and CNN-BiLSTM with the true synthetic seismic data: (a-c) near, mid, and far-angle seismic data predicted by CNN, (d-f) difference between CNN-predicted seismic data and true synthetic seismic data, (g-i) near, mid, and far-angle seismic data predicted by CNN-BiLSTM, and (j-l) difference between CNN-BiLSTM-predicted seismic data and true synthetic seismic data.

Loss function

The loss function designed in this study is used to optimize the seismic inversion model, which combines physics-informed constraints (also known as model constraints) with constraints from numerically trained neural network methods (also known as numerical constraints). This loss function integrates data fitting loss with smoothness constraints to ensure that the inversion results both conform to observed data and maintain reasonable spatial variation characteristics.

First, the data fitting loss is calculated by comparing seismic data predicted by model (physical) constraints and numerically trained neural networks with real seismic data. The seismic data fitting loss $L_{\text{data, Model constraint}}$ for the PINN model is defined as follows:

$$L_{\text{data, Model constraint}} = \frac{1}{N}\sum_{i=1}^{N} \left(\hat{y}_{\text{seis, Model constraint}}(i) - y_{\text{seis}}(i)\right)^2, \tag{8}$$

where $\hat{y}_{\text{seis, Model constraint}}$ represents the seismic data predicted by the PINN model, whereas y_{seis} is the real seismic data. Similarly, the seismic data fitting loss $L_{\text{data, Numerical constraint}}$ is calculated as follows:

$$L_{\text{data, Numerical constraint}} = \frac{1}{N}\sum_{i=1}^{N} \left(\hat{y}_{\text{seis, Numerical constraint}}(i) - y_{\text{seis}}(i)\right)^2, \tag{9}$$

where, $\hat{y}_{\text{seis, Numerical constraint}}$ represents the seismic data predicted by the neural network.

To improve the stability of inversion results, the model introduces a gradient smoothness loss L_{grad}, which is used to constrain abrupt changes in lateral gradients. Its calculation formula is as follows:

$$L_{\text{grad}} = \frac{1}{N}\sum_{i=1}^{N} \left(\hat{y}(i, j+1) - \hat{y}(i, j)\right)^2. \tag{10}$$

In addition, a second-order gradient smoothness loss $L_{\text{second grad}}$ is introduced to further suppress high-frequency oscillations in the inversion results, which is defined as follows:

$$L_{\text{second grad}} = \frac{1}{N}\sum_{i=1}^{N} \left(\hat{y}(i, j+2) - \widehat{2y}(i, j+1) + \hat{y}(i, j)\right)^2. \tag{11}$$

The final total loss function L_{total} is represented by the weighted sum of data fitting loss and smoothness constraint terms:

$$\begin{aligned} L_{\text{total}} = {} & \lambda_{\text{Model constraint}} \cdot L_{\text{data, Model constraint}} \\ & + \lambda_{\text{Numerical constraint}} \cdot L_{\text{data, Numerical constraint}}, \\ & + \lambda_{\text{grad}} \cdot \left(L_{\text{grad}} + L_{\text{second grad}}\right) \end{aligned} \tag{12}$$

where $\lambda_{\text{Model constraint}}$, $\lambda_{\text{Numerical constraint}}$, and λ_{grad} are the weights for model constraint loss, numerical constraint loss, and gradient constraint terms, respectively. By adjusting these weight parameters, a balance can be achieved between model constraints, numerical constraints, and smoothness, thereby improving the accuracy and stability of inversion results.

To further explore the tradeoff between inversion resolution and stability, we also conducted an ablation study by removing all smoothing constraints $\left(L_{\text{grad}} + L_{\text{second grad}}\right)$, and the results are discussed in Figures 6–9.

Model testing

Operating environment

The experimental investigations were carried out on a hardware platform featuring a standard personal computer. This machine was equipped with an Intel i7-11800H CPU and 40 GB of DDR4 RAM. Regarding the software setup, Windows 11 Pro served as the operating system, and the Anaconda distribution was leveraged to establish a Python 3.6-based virtual environment. TensorFlow 2.6.0 functioned as the primary DL library, augmented by supplementary packages including SciPy, Matplotlib, NumPy, Math, and Wiggle. Code development and execution were performed using the Visual Studio Code software. To guarantee the consistency and validity of the experimental findings, all trials were executed within this standardized computing infrastructure.

Training performance

In the training of PINNs, the relative weights between model-driven loss terms (physical constraints) and numerical-driven loss terms (data fitting) are crucial for the quality of inversion results. Reasonable weight allocation can effectively balance adherence to physical laws and the degree of fitting to observed data. In the early exploration phase of this study, systematic testing and evaluation were conducted on various weight combinations (detailed comparisons are not shown here due to space limitations). Results indicate that when the model-driven loss weight is set to 0.3 and the numerical-driven loss weight is set to 0.7, under the specific model and data conditions of this study, the most balanced and high-quality inversion results can be obtained, including optimal structural detail recovery and noise suppression effects. Therefore, subsequent main result presentations and analyses all adopt this optimized weight configuration (i.e., model 0.3 + numerical 0.7).

Figure 10 shows the training loss convergence curves of CNN-PINN (Figure 10a) and CNN-BiLSTM-PINN (Figure 10b) models under mixed model and numerical driving conditions (with weights of model 0.3 + numerical 0.7). As can be seen from the figure, both models exhibit good convergence characteristics, with training losses rapidly decreasing within the initial few epochs and subsequently tending to stabilize, ultimately reaching relatively low loss values, indicating that both models have been effectively trained. Specifically, the loss reduction rate for both models significantly slows down after approximately 2–3 epochs and converges to below 0.001 within 10 epochs. However, combined with the training time comparison data for different methods in Table 1, it is evident that although both models can effectively converge, there are significant differences in computational efficiency. Under the same mixed model and numerical driving conditions, the CNN-PINN model takes 288.63 s for training, whereas the training time for the CNN-BiLSTM-PINN model significantly increases to 468.33 s, approximately 1.62 times that of the former. This time difference is mainly due to the introduction of the BiLSTM layer in the CNN-BiLSTM-PINN model, which increases the complexity and computational burden of the model. Considering convergence and training efficiency, although CNN-PINN and CNN-BiLSTM-PINN models can achieve good convergence effects under mixed driving, the CNN-BiLSTM-PINN model requires a significantly higher training time cost. Therefore, when selecting a model, a trade-off needs to be made between the potential improvement in model expressiveness (BiLSTM is typically introduced to better capture sequence dependencies) and computational

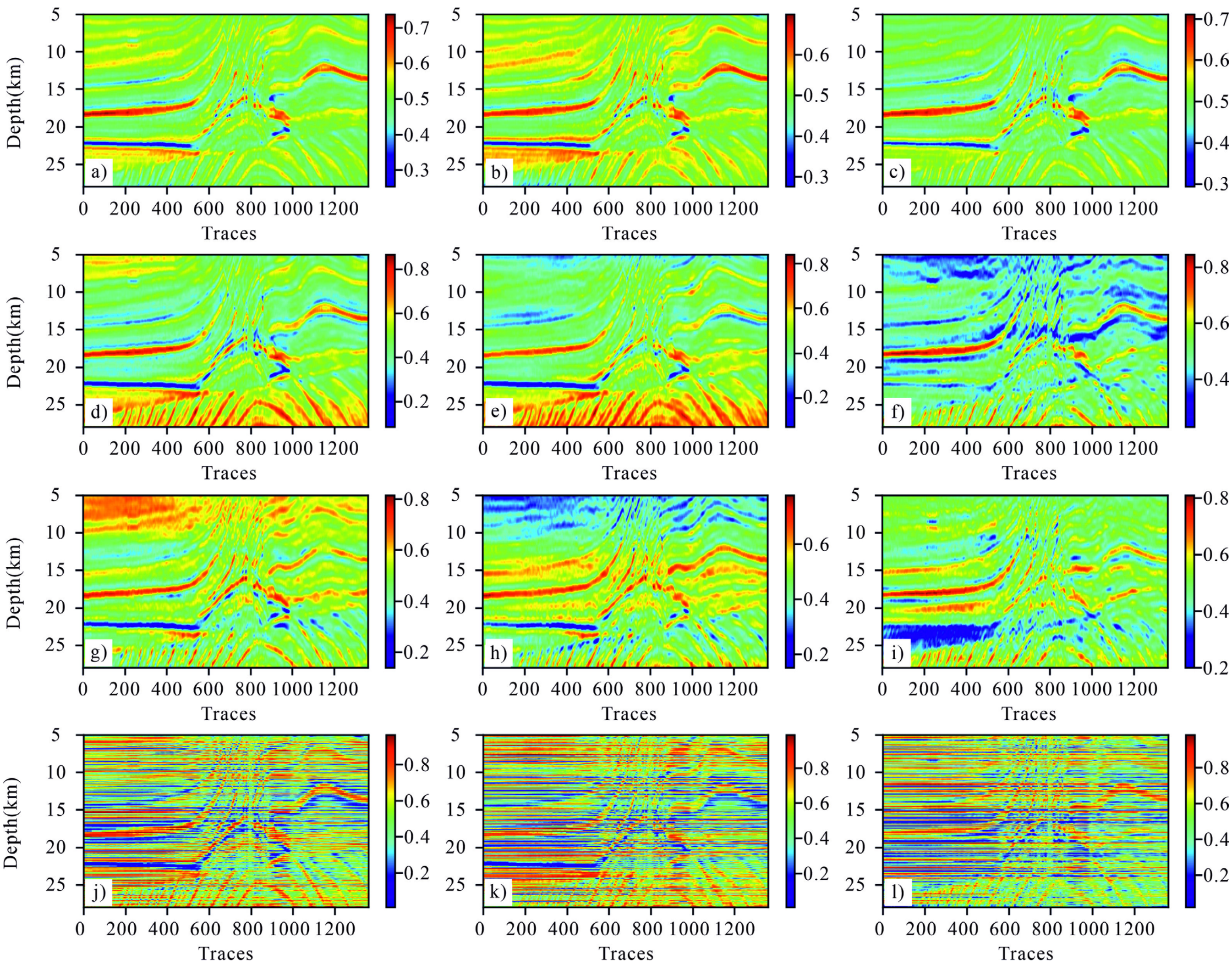


Figure 6. Comparison of model- and numerical-mixed-drive CNN-PINN inversion results: (a-c) λ, μ, and ρ from pure model-driven PINN inversion, (d-f) λ, μ, and ρ from model $\times$ 0.3 + numerical $\times$ 0.7 mixed-drive PINN inversion, (g-i) λ, μ, and ρ from pure numerical-driven PINN inversion, and (j-l) λ, μ, and ρ from mixed-drive PINN inversion without any smoothing constraint.

resource consumption. If the application scenario has high requirements for training efficiency, the relatively simpler CNN-PINN model may be more advantageous.

Figure 6 comparatively displays the inverted models of λ, μ, and ρ obtained using four different CNNPINN driving strategies (pure model-driven, pure numerical-driven, mixed model- and numerical-driven with smoothing, and mixed drive without any smoothing constraints), with the true Marmousi model (Figure 3a–3c) as the benchmark.

The pure model-driven inversion (Figure 6a–6c) produces overly smooth results for all three parameters, failing to capture important structural details and sharp impedance interfaces inherent in the true model. In contrast, the pure numerical-driven approach (Figure 6g–6i), although improving the resolution of some fine features to a certain extent compared to the model-driven approach, introduces a large amount of high-frequency noise and inversion artifacts in the inversion results for all parameters, significantly reducing the quality of shear modulus (Figure 6h) and density (Figure 6i) reconstruction.

In comparison, the mixed-driving strategy proposed in this paper, which adopts a weighted combination of model constraints and numerical constraints ($0.3 \times \text{model} + 0.7 \times \text{numerical}$; Figure 6d–6f), demonstrates significantly superior performance. Specifically, the recovered λ (Figure 6d) shows greatly improved structural clarity and sharper interfaces compared to Figure 6a while essentially eliminating the noise interference prevalent in Figure 6g. Similarly, the shear modulus inversion (Figure 6e) also demonstrates clear advantages, effectively delineating complex strata and fault structures that are unclear in Figure 6b and corrupted by noise in Figure 6h. Additionally, the density model obtained through mixed driving (Figure 6f) achieves a convincing balance: it successfully resolves key density variations and layered structures missing in the overly smooth model in Figure 6c while greatly reducing artifacts compared to the pure numerical results (Figure 6i).

To further evaluate the impact of regularization, we additionally tested the mixed-drive approach without any smoothing constraints (Figure 6j–6l). Removing this constraint slightly enhances the resolution of some fine features but also leads to increased high-frequency noise and local artifacts, particularly evident in the deeper sections of the shear modulus and density models (Figure 6k and 6l). This comparison highlights the importance of the smoothing term in balancing resolution enhancement and artifact suppression.

Overall, the results obtained through the mixed CNN-PINN method with smoothing (Figure 6d–6f) demonstrate the highest fidelity to the benchmark model across all three parameters. This method effectively uses the detailed information contained in numerical data while benefiting from the regularization effect of model constraints and smoothing, resulting in a more accurate and geologically reasonable subsurface model.

Figure 7 shows a comparison of inversion results for λ, μ, and ρ of the Marmousi2 model using the CNN-BiLSTM-PINN model under different driving modes. Figure 7a–7c presents the inversion results with pure model-driven constraints (physical equation constraints only), Figure 7d–7f shows the results with mixed model and numerical driving (weights of model 0.3 + numerical 0.7), Figure 7g–7i displays the results with pure numerical-driven constraints (data fitting only), and Figure 7j–7l illustrates the results of the mixed-drive inversion without any smoothing constraints, added to assess the impact of this regularization term on the inversion performance.

Comparative analysis reveals the following: the pure model-driven method (Figure 7a–7c) can roughly recover the main structural morphology of the model, such as anticline structures and major horizons. However, the inversion results are generally smooth, lacking fine structural details, especially with lower resolution in fault and thin-layer areas, and the accuracy of parameter values is relatively limited. This indicates that relying solely on physical equation constraints is insufficient to accurately characterize complex subsurface media.

The pure numerical-driven method (Figure 7g–7i) attempts to fit the observed data to the maximum extent, thus potentially exhibiting higher resolution in some local areas. However, there are widespread, obvious noise and artifacts in the inversion results, especially in the shear modulus and density sections (Figure 7h–7i), with poor continuity and stability of structures. This reflects the ill-posedness of the seismic inversion problem itself; without physical constraints, it is easy to overfit data noise.

The mixed model and numerical driving method (Figure 7d–7f), adopting a mixed strategy with a model driving weight of 0.3 and a numerical driving weight of 0.7, performs best for all three parameters (λ, μ, ρ). Compared with the true model, this method not only accurately reconstructs the main structural features (such as anticline morphology and fault locations) but also significantly enhances the resolution and detail depiction capability of the inversion sections, with clear horizon boundaries and parameter value distributions closer to the actual situation. Meanwhile, compared to pure numerical driving, mixed driving effectively suppresses noise and artifacts, making the inversion results more stable and geologically reasonable.

For the mixed-drive inversion without any smoothing constraints (Figure 7j–7l), some finescale structural details are further enhanced; however, this comes at the cost of increased high-frequency noise and local inversion artifacts, particularly in the deeper sections of shear modulus and density models. This comparison highlights the crucial role of the smoothing term in balancing resolution enhancement and artifact suppression.

The comprehensive comparison shows that for the CNN-BiLSTM-PINN model, the mixed model and numerical driving strategy with smoothing (model 0.3 + numerical 0.7) can effectively combine the global regularity of physical constraints with the detail depiction capability of numerical fitting. By setting appropriate weights, the accuracy and stability of seismic inversion can be significantly improved, obtaining inversion results superior to either pure model-driven or pure numerical-driven approaches and closer to the true subsurface medium parameter distribution.

Figure 8 aims to further evaluate and validate the quality of model parameters obtained from CNN-PINN inversion under different driving strategies in Figure 7through forward simulation. The figure shows synthetic seismic record sections generated using four different inversion results: pure model-driven, model 0.3 + numerical 0.7 mixed-driving, pure numerical-driven, and mixed-driving without any smoothing constraints. The evaluation criterion is the similarity between these synthetic records and the actual observed seismic data (Figure 3d–3f).

Synthetic records generated based on pure model-driven inversion results (Figure 8a–8c) can reproduce the main reflection phase axes and structural outlines, showing certain overall structural morphology. However, because their corresponding inversion models are relatively smooth (Figure 7a–7c), the generated synthetic records may not be rich enough in detail, and the continuity of phase axes and wave group characteristics may deviate from the actual data, especially with insufficiently refined responses in fault and complex structural areas.

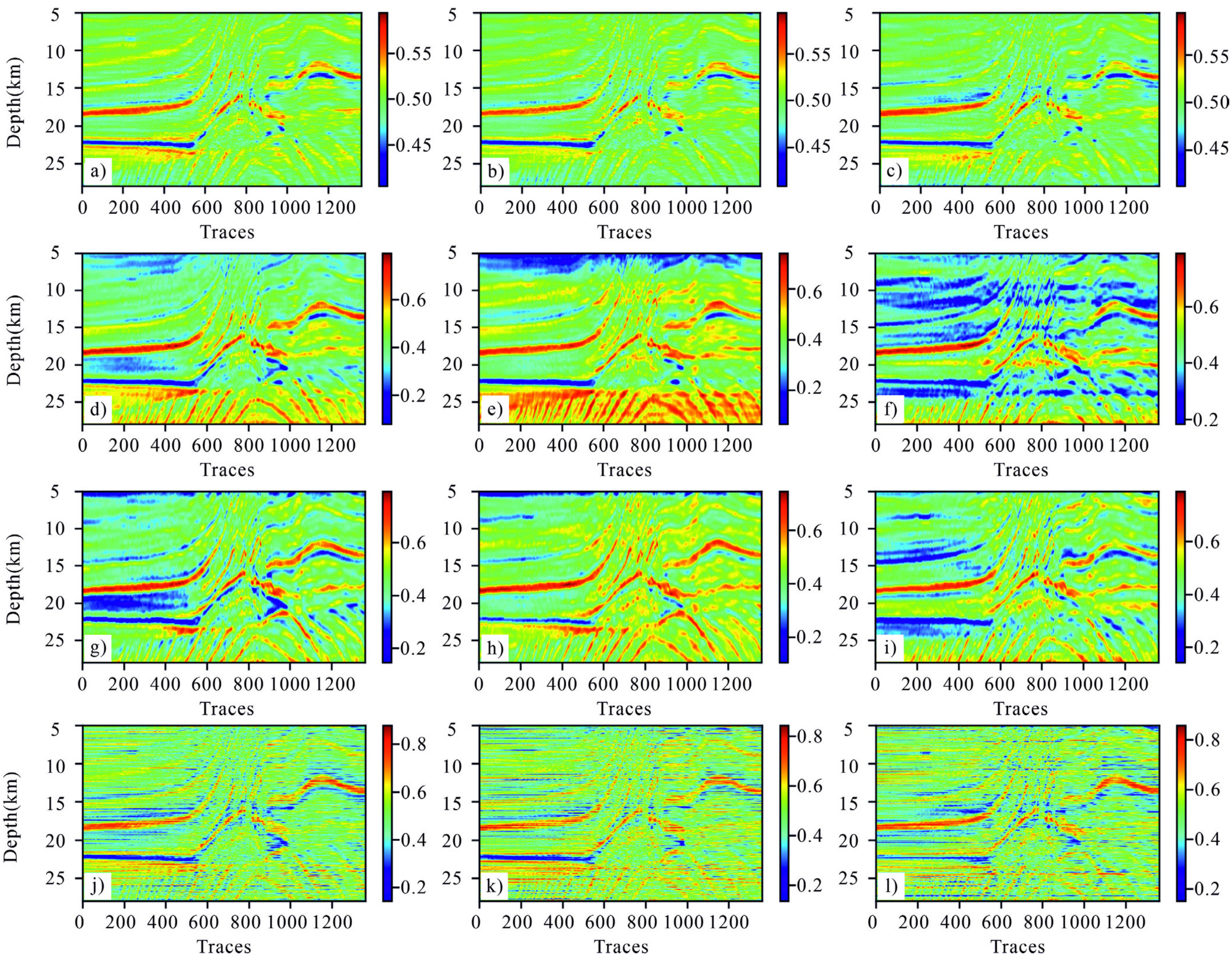


Figure 7. Comparison of model- and numerical-mixed-drive CNN-BiLSTM-PINN inversion results: (a-c) λ, μ, and ρ from pure model-driven PINN inversion, (d-f) λ, μ, and ρ from model $\times$ 0.3 + numerical $\times$ 0.7 mixed-drive PINN inversion, (g-i) λ, μ, and ρ from pure numerical-driven PINN inversion, and (j-l) λ, μ, and ρ from mixed-drive PINN inversion without any smoothing constraint.

Synthetic records generated from pure numerical-driven inversion results (Figure 8g–8i) attempt to fit the observed data to the maximum extent. Although local details may be fitted, due to the presence of more noise and artifacts in the inversion model itself (Figure 7g–7i), the generated synthetic records have a lower overall signal-to-noise ratio, with obvious chaotic reflections and discontinuous phase axes, failing to stably and reliably reflect the true underground wavefield propagation characteristics.

Synthetic records generated based on model 0.3 + numerical 0.7 mixed-driving inversion results (Figure 8d–8f) are visually closest to the expected high-quality actual seismic records. The sections are clear, with continuous and stable reflection phase axes, accurate structural detail (such as faults and folds) responses, and effectively suppressed background noise. This indicates that their corresponding inversion models (Figure 7d–7f) can best balance physical law constraints and data fitting, thereby producing synthetic seismic responses closest to the real situation.

For the mixed-driving inversion without any smoothing constraints (Figure 8j–8l), some fine reflections appear slightly sharper compared to the smoothed mixed-drive results, but this comes at the cost of significantly increased high-frequency noise and discontinuity of phase axes, particularly in deeper sections. These results highlight the crucial role of the smoothing term in ensuring the geologic plausibility and stability of forward-simulated wavefields.

By comparing forward simulation records of different inversion results, it can be concluded that the inversion model obtained by the CNN-PINN method with mixed model and numerical driving (weights of model 0.3 + numerical 0.7) generates synthetic seismic data that best matches the characteristics of actual observed data. This further validates the superiority of this mixed-driving strategy in the

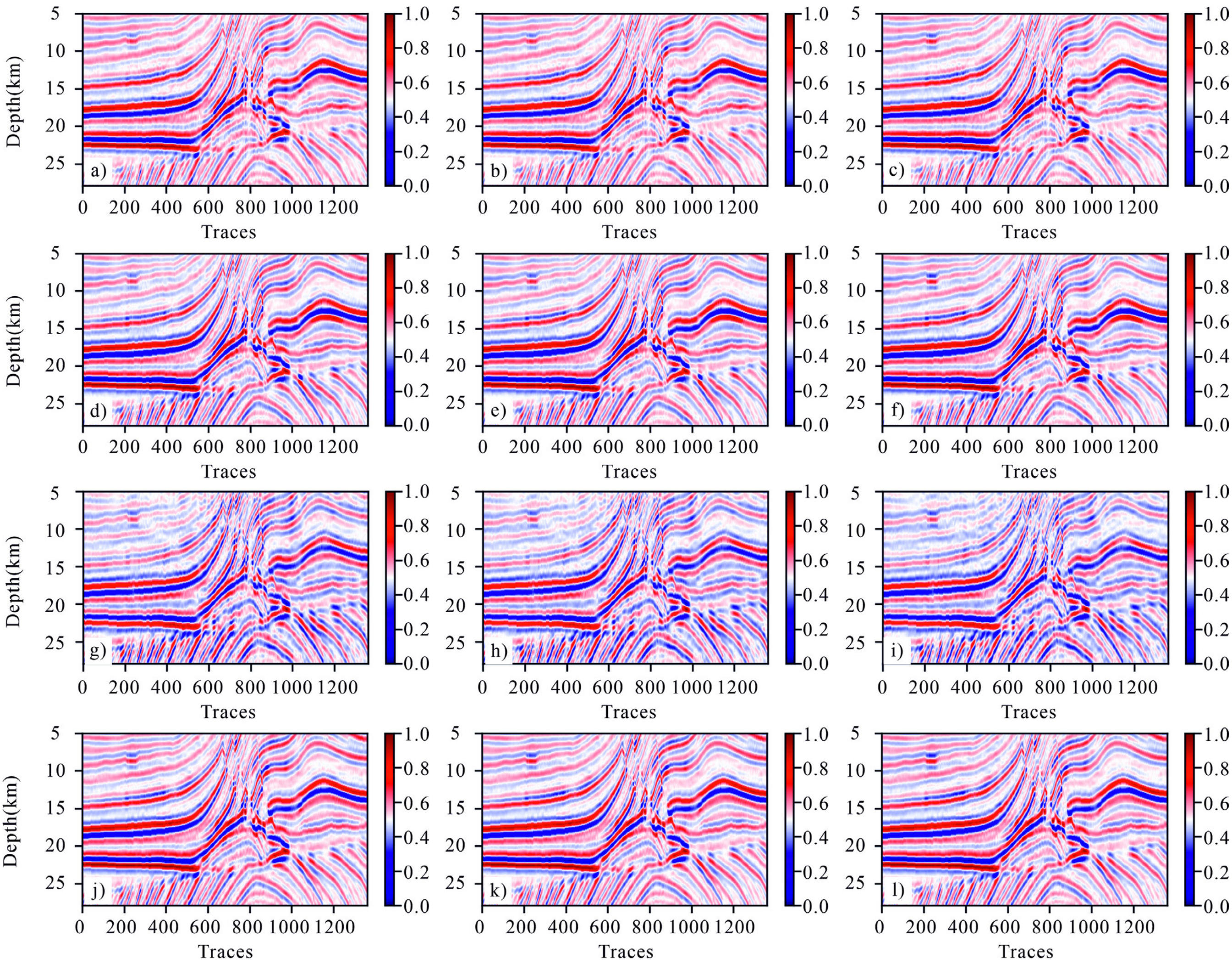


Figure 8. Comparison of CNN-PINN forward modeling results with different model- and numerical-mixed-drive weights: (a-c) forward modeling results from pure model-driven PINN inversion, (d-f) forward modeling results from model × 0.3 + numerical × 0.7 mixed-drive PINN inversion, (g-i) forward modeling results from pure numerical-driven PINN inversion, and (j-l) forward modeling results from mixed-drive PINN inversion without any smoothing constraint.

inversion process, proving that its inversion results not only are closer to the true model in the parameter domain (Figure 7) but also can better reproduce the observed seismic wavefield response in the data domain, thereby enhancing the reliability and accuracy of the inversion results.

Figure 9 shows synthetic seismic records generated by forward simulation using inversion model

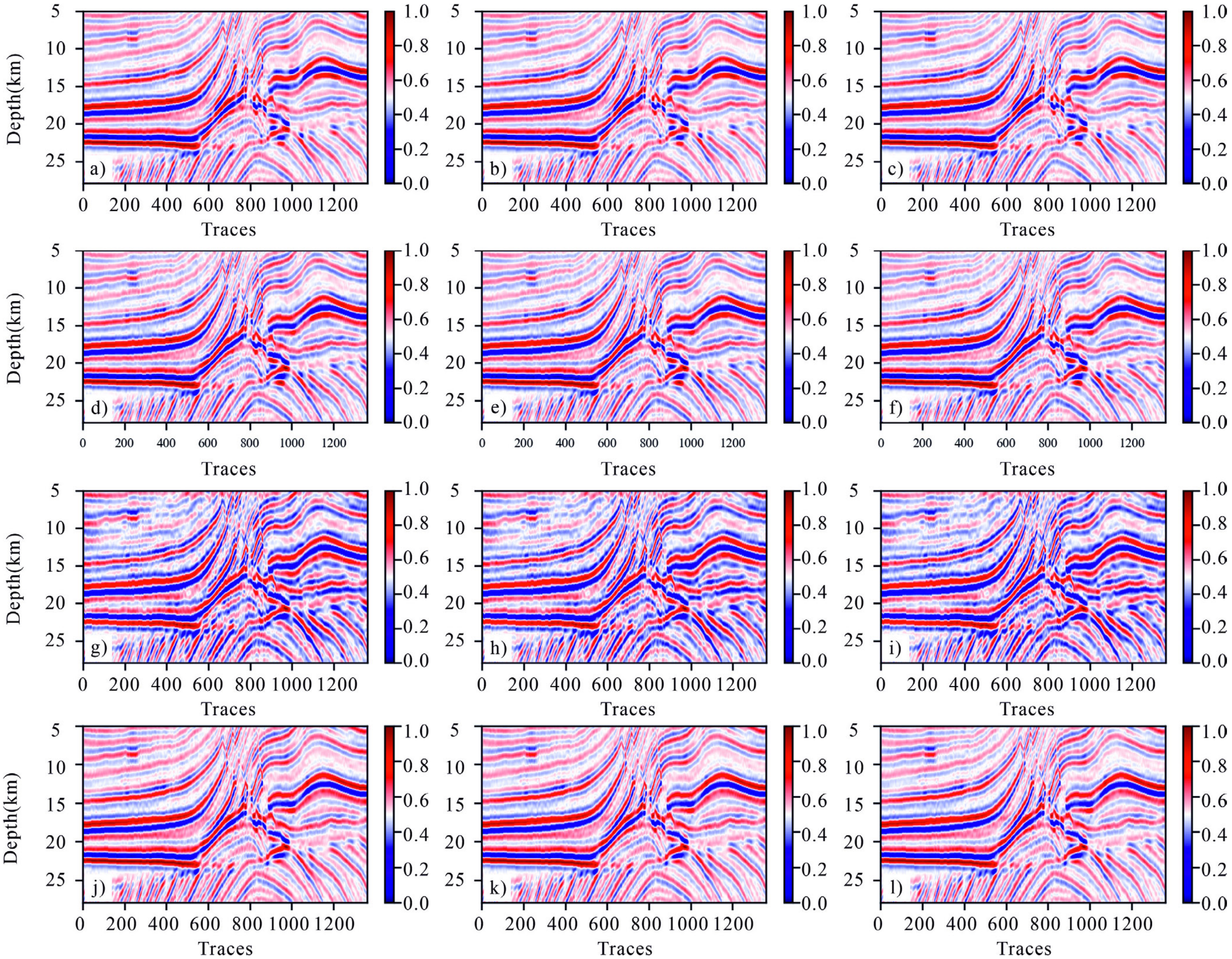


Figure 9. Comparison of CNN-BiLSTM-PINN forward modeling results with different model- and numerical-mixed-drive weights: (a-c) forward modeling results from pure model-driven PINN inversion, (d-f) forward modeling results from model × 0.3 + numerical × 0.7 mixed-drive PINN inversion, (g-i) forward modeling results from pure numerical-driven PINN inversion, and (j-l) forward modeling results from mixed-drive PINN inversion without any smoothing constraint.

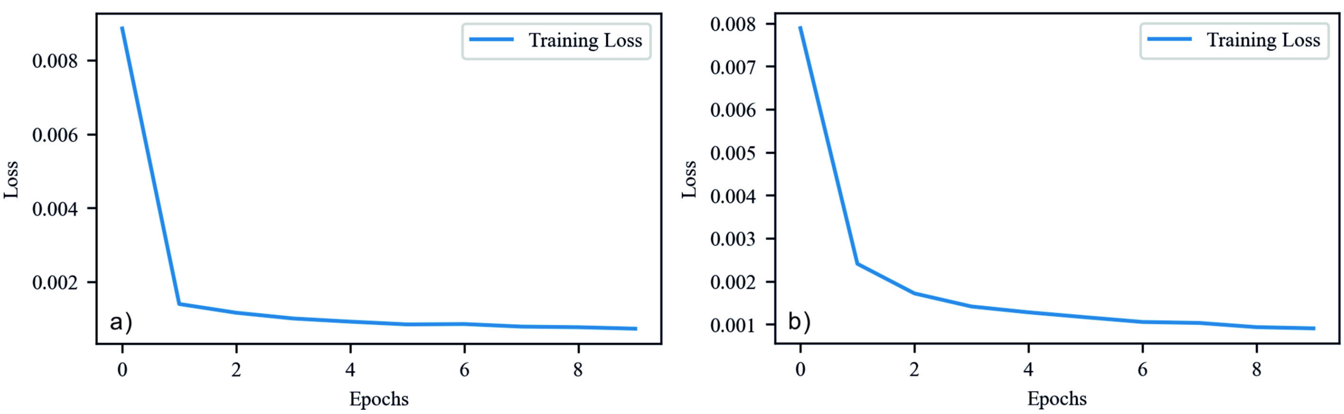


Figure 10. Comparison of convergence curves between (a) CNN-PINN and (b) CNN-BiLSTM-PINN (taking model 0.3 + numerical 0.7 as an example).

parameters obtained from the CNN-BiLSTM-PINN model under different driving strategies (referring to Figure 7). The purpose of this figure isto further evaluate the quality of inversion results under different driving strategies by comparing the similarity between synthetic records and actual observed data (Figure 3d–3f).

Synthetic records generated based on pure model-driven inversion results (Figure 10a–10c) can reflect the basic outlines of underground structures and major reflection horizons. However, due to the lack of detail depiction in their corresponding inversion models, the generated synthetic records may have certain differences in wave group characteristics, fine morphology, and the continuity of phase axes compared to actual data with high signal-to-noise ratios, especially with insufficient response in complex areas such as faults and thin interbeds.

Synthetic records generated from pure numerical-driven inversion models (Figure 10g–10i), although aimed at maximizing the fitting of observed data, may have low overall quality due to the introduction of more noise and artifacts in the inversion model itself (Figure 7g–7i). The sections may contain chaotic reflections or overly fragmented phase axes that do not conform to geologic patterns, making it difficult to stably and accurately reproduce the propagation characteristics of the actual seismic wavefield.

Synthetic records generated from inversion results using the model 0.3 + numerical 0.7 mixed-driving strategy (Figure 9d–9f) visually match high-quality actual seismic records the best. These sections exhibit clear, continuous, and stable reflection phase axes, accurate structural detail responses, and low background noise levels. This strongly proves that their corresponding inversion models (Figure 7d–7f) have achieved a good balance in combining physical priors with data fitting, capable of generating synthetic data closest to the response of real subsurface media.

For the mixed-driving inversion without any smoothing constraints (Figure 9j–9l), the synthetic records show slightly sharper local reflections but suffer from noticeably higher noise levels and disrupted continuity, again confirming that the smoothing term is essential for achieving geologically meaningful forward simulations.

Through comparative analysis of forward simulation records of CNN-BiLSTM-PINN inversion results under different driving strategies, the results show that the inversion model obtained with mixed model and numerical driving (with weights of model 0.3 + numerical0.7 in this example) can generate synthetic records most consistent with the characteristics of actual seismic data. This further confirms the advantages of this mixed-driving strategy in improving the accuracy and reliability of seismic inversion models, proving that its inversion results are not only closer to true values in the model parameter domain but also better match the observed seismic responses in the data domain. This indicates that increasing resolution by reducing regularization inevitably introduces instability, which motivates future work on adaptive smoothing strategies.

Comparative analysis of network architecture performance

To evaluate the impact of different network architectures on inversion performance, this study conducted a horizontal comparison of inversion results between CNN-PINN (Figure 6) and CNN-BiLSTM-PINN (Figure 7) under different driving strategies. The parameters examined include λ, μ, and ρ.

Under pure model-driven conditions (comparing Figure 6a–6cwith Figure 7a–7c), both architectures can roughly recover the main structural outlines of the Marmousi model, but the inversion results are generally smooth, lacking fine geologic details. Although CNN-BiLSTM-PINN (Figure 7a–7c) may show slight advantages in structural continuity, indicating that BiLSTM has a certain auxiliary role for global structure, both suffer from insufficient detail depiction due to the lack of data constraints. By contrast, under pure numerical-driven conditions (comparing Figure 6g–6iwith Figure 7g–7i), the inversion results of both models exhibit significant noise and artifacts, making it difficult to form continuous structures conforming to geologic patterns, highlighting the inherent difficulties of relying solely on data fitting for inversion without physical constraints.

When adopting a mixed model and numerical driving strategy (especially comparing Figure 6d–6f with Figure 7d–7f, with model weight 0.3 and numerical weight 0.7), the performance differences between the two architectures are most evident. For λ and μ, CNN-BiLSTM-PINN (Figure 7d and 7e) provides clearer boundaries, better structural continuity, and improved resolution than CNN-PINN, particularly in fault delineation and thin-layer identification. However, for density (ρ), CNN-PINN (Figure 6f) demonstrates statistically significant advantages across all major error metrics ($p < 0.001$), as confirmed by independent t-tests (Table 2). This indicates that CNN-BiLSTM-PINN can more effectively integrate physical constraints with data information for improved λ and μ inversion; however, CNNPINN remains more robust in density estimation under the same mixed-driving strategy.

Comprehensive comparative analysis shows that although both architectures have limitations in pure driving modes, CNN-BiLSTM-PINN demonstrates clear advantages in λ and μ inversion under mixed-model and numerical driving conditions, whereas CNN-PINN shows superior performance for ρ. This is mainly attributed to the BiLSTM structure's ability to capture spatial (or temporal) sequence dependencies in seismic data, effectively complementing CNN's local feature extraction, thereby enhancing resolution and continuity for certain parameters. Therefore, for complex inversion tasks, a CNN-BiLSTM-PINN architecture with a mixed-driving strategy is preferred for λ and μ, whereas CNN-PINN offers better density inversion performance.

Statistical significance analysis

To quantitatively validate the observed differences between CNN-PINN and CNN-BiLSTM-PINN, we conducted independent t-tests on 10 repeated experiments for all key evaluation metrics (MAE, MSE, RMSE, and R^2) of the three inverted parameters (λ, μ, and ρ). The results (Table 2) indicate that for density (ρ), CNN-PINN significantly outperforms CNN-BiLSTM-PINN across all major error metrics ($p < 0.001$). By contrast, for λ and μ, no statistically significant differences were observed between the two architectures ($p > 0.05$). These findings suggest that CNN-BiLSTM-PINN provides clearer boundaries and better structural continuity for λ and μ inversion; however, CNN-PINN is more robust in density estimation under the same mixed-driving strategy.

Additionally, negative R^2 values observed in some density inversion results are attributed to the low sensitivity of seismic data to density and the absence of strong prior model constraints in this study. Despite this challenge, CNN-PINN still achieves statistically superior performance for density inversion, reflecting the robustness and practical applicability of the proposed method.

Validation with real data

Overview of the study area

This research examines the X Field, positioned centrally within the Jin-Jia–Zheng-Lizhuang–Fan-Jia nose-like structural zone, which lies inside the Boxing Sag of the Dongying Depression. This structural zone forms part of a highly intricate hydrocarbon accumulation system, hosting reservoirs of varying depths and lithologies. Due to its abundant hydrocarbon potential, this region offers an excellent geologic framework for investigating CO_2-enhanced oil recovery (EOR) techniques.

The Boxing Sag, a subsidiary depression in the western sector of the Dongying Depression, is bounded to the west by the Qingcheng Uplift along the Gaoqing Fault and is separated from the eastern Dongying Depression by the Chunhua Structure. To the south, it connects with the Luxi Uplift, whereas to the north, it transitions into the Lijin Sag. Structurally, the sag exhibits an asymmetrical half-graben configuration, with a steep northwestern margin and a more gradually sloping southeastern side, marked by northwest-directed faulting and southeastward stratigraphic overlap. The primary drivers of its depositional evolution have been tectonic activities associated with the Shicun Fault and Gaoqing Fault. During the early Paleogene, the region's subsidence center was located within the hanging wall of the Shicun Fault, but as the period progressed, it migrated westward toward the Gaoqing area. The Jin-Jia–Fan-Jia nose-like structural zone, an inherited tectonic feature, extends longitudinally from south to north, effectively dividing the sag into eastern and western

Table 2. Results of independent *t*-tests comparing CNN-PINN and CNN-BiLSTM-PINN (mean ± standard deviation [std], *t*-value, *p*-value, and significance for *p* < 0.05).

Metric	Parameter	CNN-PINN (mean ± std)	CNN-BiLSTM-PINN (mean ± std)	t-value	p-value	Significance ($p < 0.05$)
MAE	λ	0.21936 ± 0.01238	0.21644 ± 0.00452	0.663	0.51548	
MAE	μ	0.23918 ± 0.01172	0.24726 ± 0.00686	–1.785	0.09118	
MAE	ρ	0.14727 ± 0.01204	0.16821 ± 0.00759	–4.08	0.00079	Yes
MSE	λ	0.06263 ± 0.00748	0.06110 ± 0.00264	0.578	0.5707	
MSE	μ	0.07877 ± 0.00949	0.07630 ± 0.00367	0.729	0.4752	
MSE	ρ	0.03241 ± 0.00442	0.04254 ± 0.00329	–5.521	0.00003	Yes
RMSE	λ	0.24981 ± 0.01506	0.24713 ± 0.00534	0.502	0.62157	
RMSE	μ	0.28017 ± 0.01667	0.27615 ± 0.00663	0.673	0.50945	
RMSE	ρ	0.17960 ± 0.01235	0.20611 ± 0.00795	–5.412	0.00004	Yes
R^2	λ	0.22281 ± 0.02983	0.24177 ± 0.02381	0.578	0.57069	
R^2	μ	0.25645 ± 0.08957	0.27979 ± 0.03461	–0.729	0.47523	
R^2	ρ	–0.12703 ± 0.15353	–0.47941 ± 0.11439	5.521	0.00003	Yes

MAE = mean absolute error, MSE = mean squared error, RMSE = root mean squared error.

sections. This structural zone has played a fundamental role in controlling sediment deposition and hydrocarbon reservoir formation.

According to drilling data, the stratigraphic sequence in the study area consists, from top to bottom, of the Quaternary Pingyuan, Neogene Minghuazhen, and Guantao Formations, followed by the Paleogene Dongying, Shahejie, and Kongdian Formations. The Shahejie Formation is further divided into four members, designated Sha-1 through Sha-4. The fourth member (Sha-4) exhibits a progressive lithological transition, with the basal section composed of red mudstone interbedded with sandstone, gradually giving way to gray mudstone, thin-bedded sandstone, marlstone, and oil shale in the upper section. This transformation reflects a shift in depositional environments from an arid alluvial-lacustrine plain to shallow and semideep lacustrine settings under more humid climatic conditions. The present study focuses specifically on the upper submember of Sha-4, which, based on lithological and electrical properties, is further classified into the Upper Chun and Lower Chun sub-submembers.

As depicted in Figure 11, the X Field serves as a demonstration site for CO_2 capture, utilization, and storage technology, an initiative led by Sinopec Shengli Oilfield. The spatial distribution of injection and production wells within the field is systematically arranged, and the movement of injected CO_2 closely follows the structural trends, indicating favorable geologic conditions. From the commencement of CO_2 injection in 2008, the scale of operations has steadily expanded. To date, more than 350,000 cubic meters of CO_2 have been injected, resulting in an accumulated oil production exceeding 330,000 tons. The implementation of this technique has not only led to a substantial increase in oil recovery but also provided significant economic and environmental advantages. This project offers valuable insights for optimizing the development of comparable oilfields and advancing sustainable resource management practices.

Inversion performance

Figure 12 shows the comparative results of CNN-PINN and CNN-BiLSTM-PINN in elastic parameter inversion. In wells A and B, the postinjection well logs were low-pass filtered to 60 Hzto match the effective frequency band of the seismic inversion results and suppress high-frequency noise, ensuring a reliable comparison between logs and inversion results. The CO_2 injection intervals are marked in yellow. Well A serves as the CO_2 injection well, with the injected CO_2 mainly affecting the area near well B but not reaching well C, which is consistent with the available well logs. Notably, wells A and B have postinjection logging data, whereas well C does not. Therefore, the inversion results near well C cannot be directly validated, making the geologic plausibility of model predictions in this unlogged area especially critical for assessing reliability. For derived parameters ($\lambda \times \rho$ and $\mu \times \rho$), these were not independently inverted but calculated by multiplying the inverted Lamé parameters (λ, μ) with the inverted ρ point by point, providing additional indicators for characterizing elastic property variations associated with CO_2 injection.

For λ, CNN-PINN (Figure 12a) roughly captures the elastic parameter changes caused by CO_2 injection near well A but shows lower resolution at formation boundaries, especially in the transition areas around wells B and C. CNN-BiLSTM-PINN (Figure 12b) exhibits higher resolution and clearer boundary features between T6x and T7x, more accurately reflecting the CO_2 flow impact, reaching well B but not extending to well C. This is supported by the well logging data at wells A and B and aligns with geologic expectations for the unlogged well C area.

For the shear modulus (μ), CNN-PINN (Figure 12c) generally reflects the CO_2 injection impact from well A to well B, but details near well B are blurred, and the unaffected well C area shows some inversion artifacts. CNN-BiLSTM-PINN (Figure 12d) more accurately depicts the CO_2 flow range, showing higher resolution between wells A and B while maintaining the unaffected characteristics around well C, further verifying its reliability.

For density (ρ), CNN-PINN (Figure 12e) indicates the overall trend of density reduction due to CO_2 injection but lacks clarity in the injection range, particularly near well B. CNN-BiLSTM-PINN (Figure 12f) provides a clearer response, accurately showing density changes between wells A and B while maintaining the unaffected characteristics around well C.

For derived parameters ($\lambda \times \rho$ and $\mu \times \rho$), CNN-PINN (Figure 12g and 12i) reflects the general CO_2 flow trend but lacks accuracy in boundary definition. CNN-BiLSTM-PINN (Figure 12h and 12j) delineates the injection impact more clearly, accurately depicting the effect on well B while maintaining the unaffected characteristics of well C.

In summary, although CNN-PINN provides reasonable overall elastic parameter trends, it has deficiencies in resolution and boundary clarity, especially in capturing the CO_2 flow range and boundaries. In comparison, CNN-BiLSTM-PINN, with enhanced temporal information modeling, demonstrates higher resolution and boundary accuracy. It precisely captures the CO_2 flow range from well A to well B, matches the available well logs, and maintains geologic consistency in the unlogged well C area, highlighting its advantages in complex reservoir evaluation and CO_2 EOR projects.

Figure 13 shows a comparison between synthetic seismic data predicted by CNN-PINN and CNN-BiLSTM-PINN models and actual angle-dependent seismic data, including three cases: near angle, mid angle, and far angle. Through detailed analysis, it can be found that CNN-BiLSTM-PINN demonstrates significantly better capability than CNN-PINN in capturing seismic reflection characteristics at various angles.

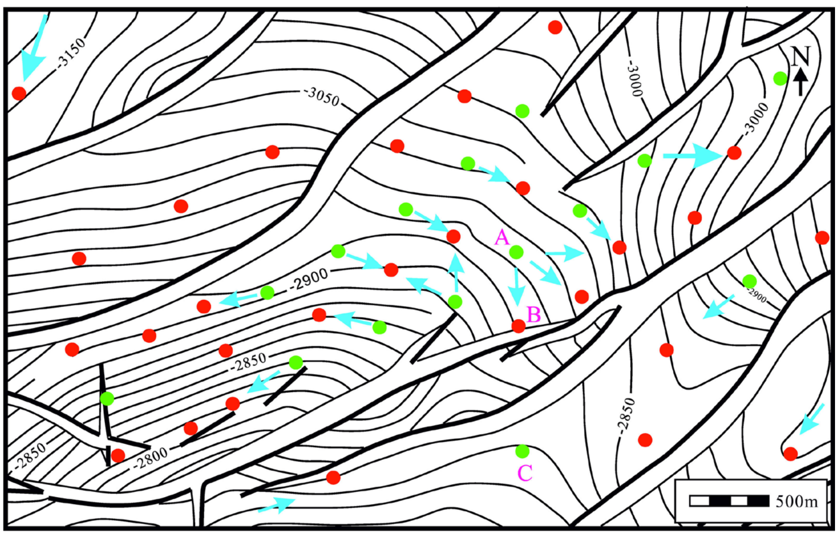
For near-angle synthetic seismic data, although the CNN-PINN result (Figure 13a) can roughly reflect the overall trend of the actual data, there are obvious differences in amplitude and phase matching, especially in local reflection layers where waveforms appear blurred and boundary reflections are not clear enough. In contrast, the near-angle result of CNN-BiLSTM-PINN (Figure 13d) shows a higher degree of consistency with the actual data, with enhanced boundary reflection characteristics and amplitude and phase closer to the actual data. The bidirectional long short-term memory network introduced in CNN-BiLSTM-PINN can better fit near-angle seismic reflection characteristics, capture temporal dependencies, and make the prediction results more accurate.

For mid-angle synthetic seismic data, the CNN-PINN result (Figure 13b) reflects the trends of major geologic layers, but there are significant mismatches in amplitude and phase in areas with large reflection coefficient variations, especially at deeper geologic interfaces. The mid-angle result of CNN-BiLSTM-PINN (Figure 13e) significantly improves the fitting with actual data. Formation interface reflection characteristics are more pronounced, with amplitude and phase better aligned with actual data, especially in deeper and high-contrast reflection areas. CNN-BiLSTM-PINN, through enhanced spatiotemporal information processing capabilities, can more accurately reconstruct complex wavefield characteristics of mid-angle seismic data.

For far-angle synthetic seismic data, although the CNN-PINN result (Figure 13c) is consistent with the overall trend of the actual data, it fails to accurately capture amplitude variations (AVO effects). Reflection continuity is poor, with local phase mismatches. The far-angle result of CNN-BiLSTM-PINN (Figure 13f) better reproduces the amplitude variation characteristics of the actual data, significantly improving the fitting of far-angle seismic reflection characteristics, such as AVO effects. Boundary amplitude, contrast, and phase consistency are notably better in the CNN-BiLSTM-PINN result. CNN-BiLSTM-PINN can more effectively handle complex wavefield changes in far-angle data, thanks to its enhanced sequence modeling capability.

The comparison of synthetic seismic data generated by CNN-PINN and CNN-BiLSTM-PINN shows that CNN-BiLSTM-PINN outperforms CNN-PINN in amplitude and phase fitting of near, mid, and far-angle seismic data. The results of CNN-BiLSTM-PINN are particularly outstanding in boundary clarity and performance in deep regions. CNN-PINN, limited by its convolutional network structure, does not perform ideally in complex reflection areas such as far-angle AVO effects and high-contrast formation boundaries. On the other hand, CNN-BiLSTM-PINN, with its enhanced temporal modeling capability from bidirectional long short-term memory networks, better captures seismic reflection characteristics, achieving a higher degree of consistency with actual seismic data. This highlights CNN-BiLSTM-PINN's stronger robustness and adaptability under complex geologic conditions.

In conclusion, CNN-BiLSTM-PINN significantly improves the quality of synthetic seismic data by introducing bidirectional long short-term memory networks, enhancing the matching with actual seismic data, especially in capturing complex reflection characteristics in mid and far-angle data. In comparison, although CNN-PINN reasonably reproduces the overall trend, it has limited ability in capturing details and specific reflection characteristics. These results highlight CNN-BiLSTM-PINN's wider applicability and higher precision in seismic inversion and complex reservoir characterization.



●Gas Well ●Oil Well (CO_2 Injection Well) A Well Name → CO_2 Flow Pathway

Figure 11. Distribution map of mainstream directions of CO_2 flooding in the study area.

Discussion

Why does CNN-BiLSTM demonstrate higher efficiency than CNN in supervised learning, yet show reduced efficiency in PINN?

Research through comparative experiments reveals differences in performance characteristics between CNN and CNN-BiLSTM models under different learning paradigms.

Experimental results show that in supervised learning scenarios, the CNN-BiLSTM model demonstrates significant efficiency advantages; despite slightly more training epochs (205 epochs compared to CNN's 195 epochs), the total training time is still reduced by approximately 12.3% (144.31 s compared to CNN's 164.64 s). This efficiency improvement is mainly attributed to the BiLSTM layer's effective modeling

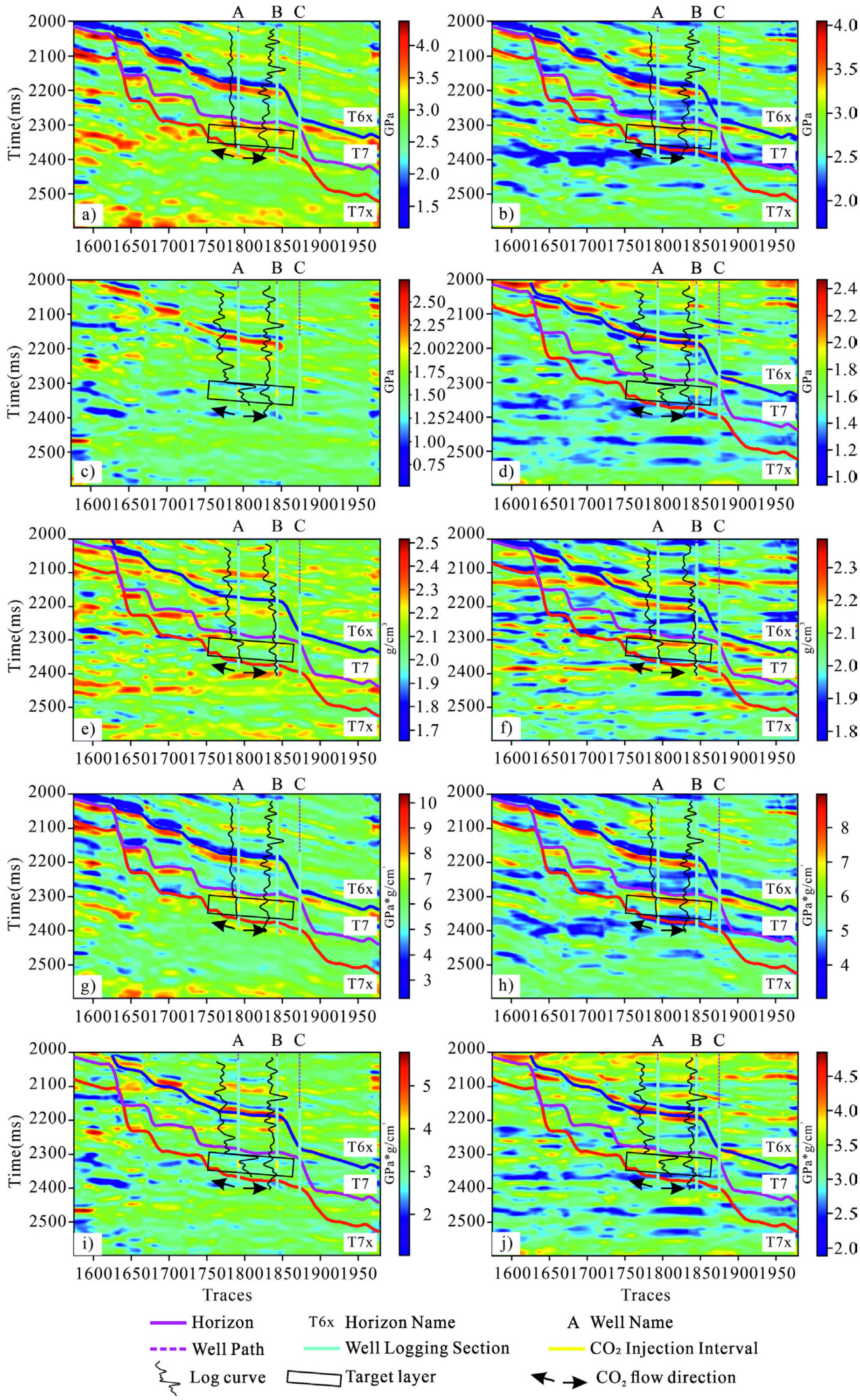


Figure 12. Comparison of inversion results between CNN-PINN and CNN-BiLSTM-PINN with 60-Hz-filtered well logs (wells A and B; no logs for well C). Well logs were acquired after CO_2 injection and CO_2 injection intervals are marked in yellow: (a) CNN-PINN: lambda (λ), (b) CNN-BiLSTM-PINN: lambda (λ), (c) CNN-PINN: mu (μ), (d) CNN-BiLSTM-PINN: mu (μ), (e) CNN-PINN: density (ρ), (f) CNN-BiLSTM-PINN: density (ρ), (g) CNN-PINN: lambda × density ($\lambda \times \rho$), (h) CNN-BiLSTM-PINN: lambda × density ($\lambda \times \rho$), (i) CNN-PINN: mu × density ($\mu \times \rho$), and (j) CNN-BiLSTM-PINN: mu × density ($\mu \times \rho$).

capability for temporal dependencies, enabling the model to more quickly identify key patterns in the data and accelerate the convergence process. However, when these model architectures are integrated into the PINN framework, performance characteristics change significantly. In the PINN environment, although the CNN-BiLSTM-PINN model exhibits smoother convergence curves, its computational overhead increases significantly: under the mixed model and numerical driving mode, its training time increases by approximately 62.3% compared to CNN-PINN (468.33 s versus 288.63 s); under pure model-driven mode, it increases by approximately 10.5% (417.2 s versus 377.67 s); and under pure numerical-driven mode, it increases by approximately 32.0% (398.65 s versus 302.07 s).

This seemingly contradictory phenomenon can be attributed to the interaction between physical constraint calculations and the BiLSTM recursive structure. In the PINN framework, derivatives of physical equations need to be calculated, and when these derivative calculations pass through the complex recursive structure of BiLSTM, they generate more complex computational graphs, significantly increasing the computational burden on the automatic differentiation system. Additionally, the combination of physical constraints with the nonlinear characteristics of BiLSTM may create a more complex loss landscape, requiring more computational resources to optimize.

This finding has important implications for the application of DL models in scientific computing: model architectures may exhibit drastically different performance characteristics under different learning paradigms, and selecting a model architecture suitable for a specific application scenario requires comprehensive consideration of multiple factors such as accuracy, stability, and computational efficiency. For applications requiring high-precision physical simulation, the stable convergence characteristics of CNN-BiLSTM-PINN may be more important; however, for applications with limited computational resources or high real-time requirements, CNN-PINN may be a more appropriate choice.

Advantages of PINN compared to traditional supervised neural network inversion

PINNs demonstrate multiple significant advantages over traditional supervised neural networks in the field of geophysical inversion. First, PINN ensures that inversion results strictly adhere to fundamental physical laws by directly incorporating physical principles (such as LMR formulas) into the neural network training process, significantly enhancing the model's interpretability and geologic reasonableness. Because physical models have been developed over decades, their generalization capabilities are widely recognized. This introduction of physical constraints effectively avoids physically unreasonable solutions that purely numerical-driven methods might produce, providing a solid theoretical foundation for inversion results. Second, PINN significantly reduces dependence on large amounts of labeled data. Traditional supervised neural networks typically require extensive high-quality labeled data pairs for training, which are often difficult to obtain in the geophysical domain. Even using synthetic data for supervised neural network training faces matching issues between synthetic and actual data. PINN achieves completely unsupervised inversion by incorporating prior knowledge provided by physical equations,

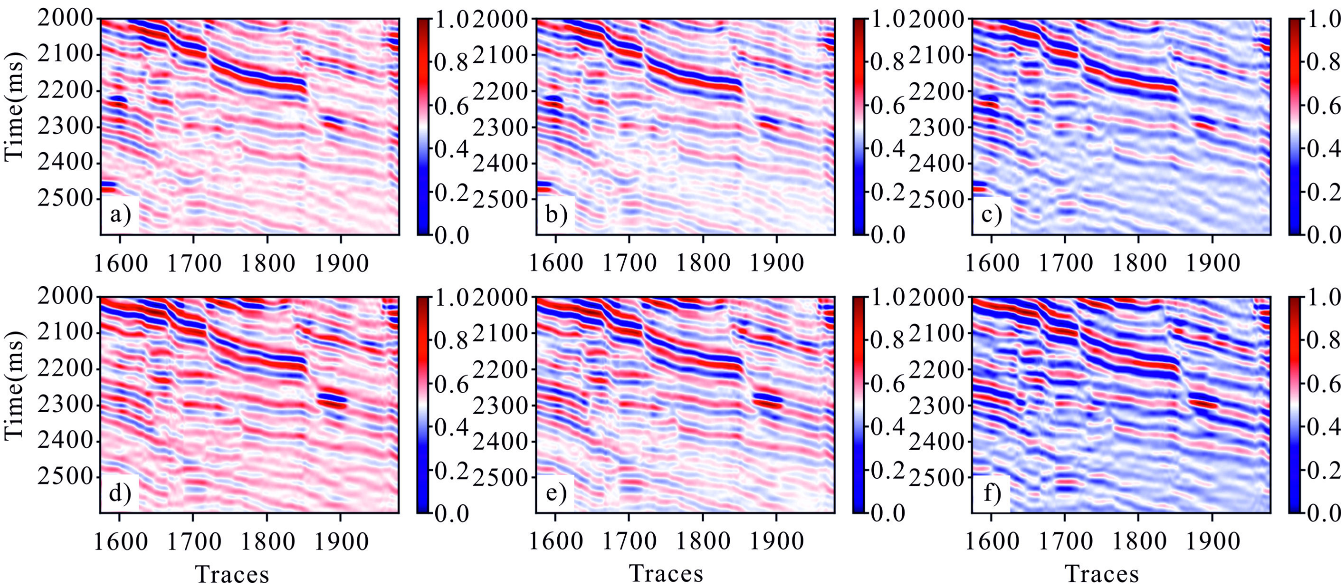


Figure 13. Comparison of seismic data predicted by CNN-PINN and CNN-BiLSTM-PINN: Synthetic seismic data versus actual angle seismic data. (a-c) Synthetic seismic data generated from CNN-PINN predictions for near, mid, and far angles, respectively. (d-f) Synthetic seismic data generated from CNN-BiLSTM-PINN predictions for near, mid, and far angles, respectively.

eliminating the need for pretraining and label data creation, effectively avoiding the risk of data mismatch, and fundamentally addressing the long-standing "few-shot learning" challenge in the geophysical field. Third, PINN demonstrates excellent stability and generalization capability in inversion results. Constrained by physical laws, its inversion results exhibit stronger stability and consistency on unseen data. This is particularly important for seismic data inversion, as the complexity and diversity of actual geologic conditions make it difficult for purely numerical-driven methods to handle all possible scenarios. Physical constraints essentially narrow the solution space, guiding the model to converge toward physically reasonable directions. Fourth, PINN possesses unique advantages in handling ill-posed inverse problems like seismic inversion. Traditional neural networks often fall into local optima or produce multiple possible solutions when facing multisolution problems; however, PINN effectively reduces the solution space through physical constraints, improving the uniqueness and reliability of inversion results, making them closer to real geologic models. Additionally, PINN exhibits stronger noise robustness. Actual seismic data typically contain various noises, and traditional neural networks easily overfit these noises, leading to unstable inversion results. Constrained by physical laws, PINN can extract essential features that conform to physical principles from noisy data, demonstrating natural resistance to noise. In conclusion, PINN demonstrates performance clearly superior to traditional supervised neural networks in seismic elastic parameter inversion by organically combining the powerful data fitting capability of DL with the prior knowledge of physical models. This hybrid-driven approach of "physics + data" represents an important development direction for geophysical inversion, providing a theoretically complete and practically efficient new solution for high-precision parameter inversion under complex geologic conditions.

Comparison with conventional methods

In comparison with traditional inversion frameworks, the proposed CNN-PINN and CNN-BiLSTM-PINN approaches exhibit distinct advantages. Purely physics-driven methods such as FWI directly minimize the misfit between observed and modeled seismic data using numerical solvers and classical optimization algorithms (e.g.,LBFGS). Although FWI maintains strong physical consistency, it often requires high-quality initial models and struggles to achieve robust convergence in geologically complex areas, such as regions with strong heterogeneity or faulted structures.

On the other hand, purely numerical-driven inversion methods based on conventional CNNs rely entirely on supervised learning using large volumes of labeled training data, which are difficult to obtain in geophysical applications. Furthermore, their lack of embedded physical constraints frequently leads to overfitting and limited generalization capability, especially when applied to unseen geologic settings.

The hybrid-driven strategy adopted in this study effectively bridges the gap between these two extremes. By embedding physical constraints (e.g.,the LMR formulation) into deep neural network training, our method achieves unsupervised inversion where the predicted results simultaneously satisfy numerical fitting and physical consistency. The use of prestack angle-domain seismic data with near-angle stacking further enhances the signal-to-noise ratio, allowing more reliable estimation of elastic parameters in challenging geologic settings. Notably, statistical tests reveal that CNN-PINN demonstrates significantly better performance in density (ρ) inversion compared to CNN-BiLSTM-PINN ($p < 0.001$), suggesting that the CNN-based architecture may retain advantages in estimating density-related parameters. This integration of physical priors and numerical-driven feature extraction enables the CNN-PINN and CNN-BiLSTM-PINN frameworks to produce inversion results that are geologically plausible and more robust to data noise and structural complexity than those obtained by either purely physics-driven or purely numerical-driven approaches.

The significance and advantages of combining two constraint methods

Whether it is pure model-driven PINN or pure numerical-driven PINN, each has its advantages and disadvantages. For example, pure model-driven PINN inverts λ with higher accuracy and more details, whereas pure numerical-driven PINN can invert more refined μ and ρ. By combining these two physical constraint methods, we can fully leverage the advantages of both approaches, simultaneously achieving high-precision inversion of λ, μ, and ρ.

Through in-depth analysis of these two driving methods, it becomes evident that they exhibit clear complementarity in physical parameter inversion. Pure model-driven PINN, by strictly adhering to physical laws such as wave equations, can more accurately capture λ propagation characteristics, performing exceptionally well in complex geologic structures. This is mainly attributed to the model-driven method directly embedding wave equations as constraints into the neural network's loss function, making the inversion results more physically reasonable and preserving more geologic structural details, especially with stronger retention capability for high-frequency components.

In contrast, pure numerical-driven PINN relies on large amounts of training data, learns through implicit physical laws in the data, and exhibits unique advantages in shear modulus and density parameter inversion. This may be because shear modulus and density parameters affect numerical simulation results differently from λ, and numerical-driven methods can better extract features of these parameters from data. Additionally, numerical-driven methods typically have stronger robustness against noise and uncertainty, providing more stable inversion results under imperfect data conditions.

The fusion of these two driving methods is a process of synergistic enhancement. Through a carefully designed weight allocation strategy, the contribution ratio of model-driven and numerical-driven components in the overall loss function can be uniformly adjusted. This global weight strategy can balance the relationship between physical constraints and data fitting while maintaining physical consistency, thereby achieving overall optimal effects in the inversion of all parameters (λ, μ, and ρ). Although weights are adjusted globally, different weight ratios will affect the inversion accuracy of various physical parameters to different degrees. For example, increasing the weight of model-driven components will strengthen the physical consistency of all parameters, whereas increasing the weight of numerical-driven components will improve the model's fitting degree to training data. By systematically exploring inversion effects under different weight ratios, a balance point can be found where λ, μ, and ρ can simultaneously achieve high inversion accuracy. This fusion strategy fully uses the advantages of both driving methods, improving data fitting capability while maintaining physical reasonableness and providing an effective approach for high-precision joint inversion of geophysical parameters. Experimental results further show that this fusion-driven PINN method achieves high-precision inversion of λ and μ, whereas CNN-PINN demonstrates superior robustness in ρ estimation under the same strategy, highlighting the complementary advantages of the two network architectures.

Method limitations and future improvement directions

Although the proposed framework demonstrates significant advantages in integrating physical and numerical constraints for seismic elastic parameter inversion, it still has several limitations. One limitation lies in the assumption of a fixed Ricker wavelet during the physical forward modeling stage. Although the Ricker wavelet is widely adopted for its simplicity and analytical tractability, it cannot fully represent the bandwidth limitations, phase distortions, and source-signature variability encountered in real seismic acquisition. This simplification may lead to modeling errors when the actual source wavelet deviates significantly from the Ricker form, particularly in complex stratigraphic settings or under strong attenuation. Incorporating more realistic or data-driven wavelet estimation methods (e.g., adaptive matching filtering) could enhance inversion fidelity in practical applications.

Extending the current 2D implementation to large-scale 3D seismic inversion also presents substantial challenges. Three-dimensional forward modeling and gradient computation greatly increase computational costs and memory requirements, especially when combining BiLSTM structures with physics-informed constraints. Moreover, 3D geologic heterogeneity and anisotropy may require more sophisticated forward models and constraint formulations to maintain inversion stability and accuracy. Efficient parallelization strategies, such as multi-GPU or multinode distributed training, Compute Unified Device Architecture-based tensor computation, and mixed-precision training, alongside network optimization techniques (e.g., sparse convolution, model pruning, and lightweight architectures), will be essential for scaling the method to large-scale 3D data sets and near-real-time applications.

Additionally, the LMR formulation used as the physical constraint in this study does not explicitly account for anisotropy or frequency-dependent dispersion effects. Although this simplification is reasonable for isotropic or weakly anisotropic formations and low- to mid-frequency prestack seismic data, it may introduce biases when applied to strongly anisotropic reservoirs or high-frequency imaging scenarios. Future work could explore anisotropic reflection approximations (e.g., Rüger's equation) or frequency-dependent reflection models to extend applicability to more complex geologic settings.

Finally, the optimal weight ratio for mixed model- and numerical-driving (0.3:0.7 in this study) may vary depending on geologic conditions and data quality, and the current framework lacks an adaptive adjustment mechanism. Future research directions include: (1) developing inverse physics-informed neural networks (iPINNs) for automatic weight optimization, (2) integrating attention mechanisms and sparse convolution to reduce computational cost while maintaining accuracy, (3) exploring multiscale physical constraint fusion to better capture geologic heterogeneity, (4) extending the method to time-lapse monitoring for dynamic reservoir characterization, and (5) incorporating uncertainty quantification to provide reliability assessments of inversion results. These improvements will help further enhance the accuracy, efficiency, and practicality of physics-informed neural network approaches in seismic inversion.

Conclusion

In this study, we proposed a CNN-BiLSTM-PINN network architecture that integrates physical and numerical constraints for seismic elastic parameter inversion. Systematic experiments and analyses demonstrate that the dual-driving strategy, combining model and numerical constraints, significantly improves the accuracy and stability of seismic elastic parameter inversion. The mixed-driving approach — specifically, a combination of 30% model-driven and 70% numerical-driven constraints — enables more accurate reconstruction of subsurface elastic parameters (λ, μ, ρ), particularly in complex geologic settings. Pure model-driven methods tend to yield overly smooth results, whereas pure numerical-driven methods often introduce excessive noise and artifacts; the mixed-driving strategy effectively balances these extremes. The inclusion of BiLSTM networks in the CNN-BiLSTM-PINN architecture enhances the capability to capture spatiotemporal dependencies in seismic data, leading to substantial improvements in inversion resolution and boundary continuity for λ and μ, as compared to CNN-PINN. However, statistical analysis indicates that CNN-PINN achieves superior performance for ρ inversion, highlighting that each architecture offers distinct strengths depending on the target parameter. In supervised learning scenarios, the CNN-BiLSTM model exhibits higher training efficiency than the CNN, reducing training time to 42.4% while achieving lower final loss. Nevertheless, within the PINN framework, the computational complexity of CNN-BiLSTM-PINN increases considerably due to the interaction between physical constraint calculations and the recursive structure of BiLSTM, resulting in a training time approximately 1.62 times that of CNN-PINN. Forward simulation validation further confirms that synthetic seismic records generated from mixed-driving CNN-BiLSTM-PINN inversion results best match actual observations, as reflected by clearer and more continuous reflection phase axes, accurate responses to structural details, and effectively suppressed background noise. Comparative analysis of network architectures reveals that although CNN-PINN and CNN-BiLSTM-PINN have limitations in pure driving modes, the latter exhibits superior boundary clarity and structural continuity for λ and μ under mixed-driving conditions, whereas CNN-PINN performs better for ρ, underscoring the complementary nature of the two architectures for different inversion objectives. When applied to the X oilfield CO_2 flooding demonstration area, the CNN-BiLSTM-PINN network demonstrates substantial advantages in mapping elastic parameters associated with CO_2 migration, more precisely delineating the range of CO_2 movement between wells, and aligning perfectly with well-log data — highlighting its value for monitoring and evaluating CO_2 EOR projects. In comparing synthetic and actual angle-dependent seismic data, CNN-BiLSTM-PINN surpasses CNN-PINN in amplitude and phase fitting, especially in capturing complex reflection features such as AVO effects at mid and far angles, attributed to its enhanced sequence modeling capabilities. The fusion of model- and numerical-driven strategies operates as a synergistic process, wherein a global weighting scheme allows for flexible adjustment of the contributions of each component to the overall loss, ensuring a balance between physical consistency and data fidelity and enabling optimal inversion performance across all parameters.

Despite the notable advantages of the proposed approach, certain limitations remain, including increased computational complexity and the absence of an adaptive weighting mechanism. Future research can address these issues and further enhance the accuracy, efficiency, and practicality of PINN-based seismic inversion by incorporating attention mechanisms and sparse convolutional structures, designing iPINNs for adaptive weight adjustment, exploring multiscale physical constraint fusion, extending applications to 3D seismic data and time-lapse monitoring, and integrating uncertainty quantification techniques.

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Data and materials availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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